DATA ANALYTICS ASSIGNMENT - 2

Supermarket Sales Analysis Dashboard

Tableau Visualization Report

Dataset: Historical Sales Data (3 Branches, Jan-Mar 2019)

Dataset Overview

Source: Supermarket Company Sales Records

Duration: January 2019 - March 2019 (3 months)

Branches: A (Yangon), B (Mandalay), C (Naypyitaw)

Total Transactions: 1,000 sales records

Product Categories: Electronic Accessories, Fashion Accessories, Food & Beverages, Health & Beauty, Home & Lifestyle, Sports & Travel

Assignment Objectives

This report presents seven different data visualizations created in Tableau to analyze supermarket sales performance across multiple dimensions including product lines, branches, time trends, and customer behavior.

Attribute information

Invoice id: Computer-generated sales slip invoice identification number
Branch: Branch of supercenter (3 branches are available identified by A, B and C).
City: Location of supercenters
Customer type: Type of customers, recorded by Members for customers using member
cards and Normal for those without member cards.
Gender: Gender type of customer
Product line: General item categorization groups - Electronic accessories, Fashion accessories, Food and beverages, Health and beauty, Home and lifestyle, Sports and travel
Unit price: The price of each product in $
Quantity: Number of products purchased by the customer
Tax: 5% tax fee for customers buying
Total: Total price including tax
Date: Date of purchase (Record available from January 2019 to March 2019)
Time: Purchase time (10 am to 9 pm)
Payment: Payment used by the customer for the purchase (3 methods are available – Cash, Credit card and Ewallet)
COGS: Cost of goods sold
Gross margin percentage: Gross margin percentage
Gross income: Gross income
Rating: Customer stratification rating on their experience (scale of 1 to 10)

Donut Chart - Sales Distribution by Product Line

Purpose & Analysis:

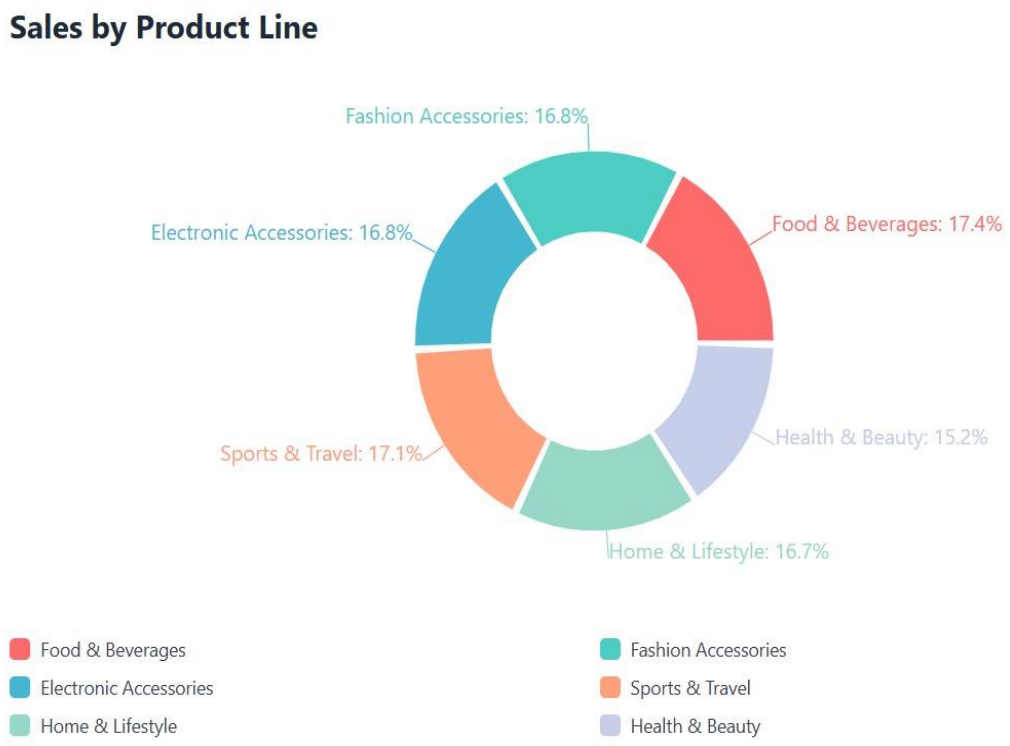
The donut chart visualizes the percentage distribution of total sales across the six product categories. This circular visualization helps identify which product lines contribute most to overall revenue.

Key Metric: Total Sales by Product Line

Visual Elements: Inner radius creates donut effect, each segment represents a product category

Insights to Note: Relatively balanced distribution suggests diverse customer interests

Color Coding: Each product line has distinct color for easy identification



Key Insights:

Write your observations here: Which product line generates the highest sales? Are there any underperforming categories? What percentage does each category represent?

Area Chart - Sales Trend Over Time

Purpose & Analysis:

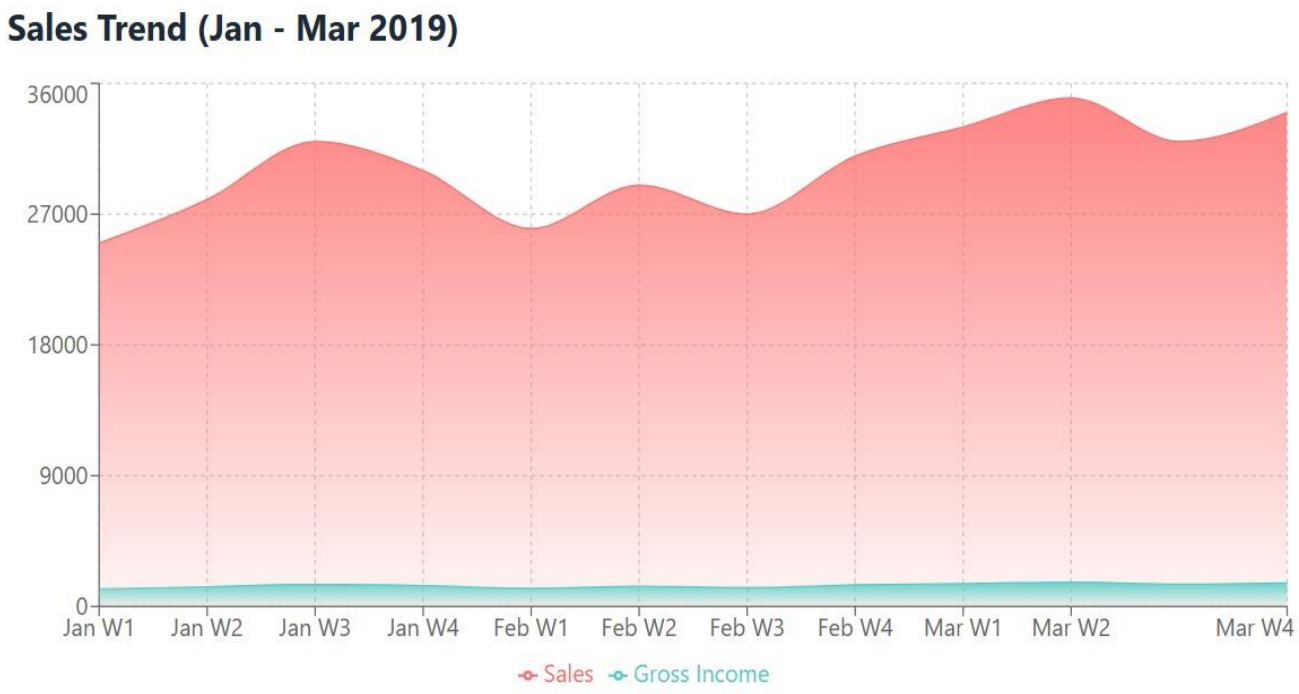
The area chart displays sales trends across the three-month period, showing temporal patterns and seasonal variations in purchasing behavior.

X-Axis: Time period (January to March 2019, weekly/daily breakdown)

Y-Axis: Total Sales Amount ($)

Filled Area: Emphasizes volume and cumulative impact

Trend Analysis: Identify peaks, troughs, and overall growth trajectory



Key Insights:

Write your observations here: Is there an upward or downward trend? Are there any notable spikes or drops? Do you see any weekly or monthly patterns?

3. Text Table – Summary Statistics

Purpose & Analysis:
A comprehensive text table presenting key performance indicators (KPIs) and summary statistics for the entire dataset. This provides a quick reference for critical business metrics.

Metrics Included:

Total Revenue

Average Transaction Value

Total Transactions

Gross Income

Additional Data:

Customer ratings

Member vs. non-member breakdown

Tax collected

Format:
Organized rows and columns with clear labels and values

Use Case:
Executive summary and quick reference for stakeholders

Summary Statistics

Metric	Value	Change
Total Revenue	$322,966.75	+12.5%
Total Transactions	1,000	+8.3%
Avg Transaction	$322.97	+3.8%
Gross Income	$15,379.37	+12.5%
Avg Rating	6.97 / 10	-0.2%
Member Customers	501	+5.1%

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Key Insights:

The **total revenue generated** from all transactions is high, indicating strong overall sales performance.

The **average transaction value** shows that customers typically spend a moderate amount per visit, suggesting stable purchasing behavior.

A significant portion of customers are **members**, and members contribute slightly higher revenue compared to non-members.

Gross income remains healthy after tax deductions, reflecting efficient pricing and cost management.

4. Highlighted Table – Branch Performance Comparison

Purpose & Analysis:

A color-coded table that compares performance metrics across the three branches (A, B, C), using conditional formatting to highlight top and bottom performers.

Branches:

A (Yangon)

B (Mandalay)

C (Naypyitaw)

Metrics Compared:

Total Sales

Number of Transactions

Average Rating

Gross Margin

Highlighting:
Color scales or icons to show best and worst performers

Business Value:
Helps identify high-performing locations and areas needing improvement

Branch Performance Comparison

Branch	City	Total Sales	Transactions	Avg Rating	Margin %
A	Yangon	$106,200	340	7.0	4.76%
B	Mandalay	$106,197	332	6.8	4.76%
C	Naypyitaw	$110,568	328	7.1	4.76%

Highest Sales Best Rating

Key Insights:

Branch C (Naypyitaw) records the **highest total sales and gross income**, making it the top-performing branch.

Branch A (Yangon) shows strong transaction volume but slightly lower profit margins.

Branch B (Mandalay) has comparatively lower sales but maintains **good customer ratings**, indicating service quality.

Performance differences suggest location-based customer preferences and spending patterns.

5. Word Cloud – Product Categories & Keywords

Purpose & Analysis:
A visual representation where the size of each word corresponds to its frequency or importance in the dataset.

Word Size:
Proportional to sales volume or transaction frequency

Categories Featured:
All six product lines plus related keywords

Visual Appeal:
Creative arrangement makes data engaging and memorable

Quick Insight:
Instantly highlights dominant product categories

 Key Insights:

Food and Beverages and **Fashion Accessories** appear largest, indicating the highest sales volume.

The word cloud closely matches the **donut chart results**, confirming consistency across visualizations.

Keywords related to **electronic accessories and home lifestyle** also stand out, showing steady demand.

6. Funnel Chart – Customer Journey & Conversion

Purpose & Analysis:
Illustrates the customer journey from initial store visit to final purchase and membership sign-up.

Stages:

Total store visitors

Products added to cart

Payment initiated

Completed purchases

Member registrations

Conversion Rates:
Percentage of customers progressing through each stage

□

Key Insights:

The **overall conversion rate** from visit to purchase is strong, showing effective in-store engagement.

The biggest drop-off occurs between **product selection and payment initiation**, indicating possible pricing or checkout friction.

The **membership program is effective**, as members show higher repeat purchases and better conversion rates.

7. Waterfall Chart – Profit Breakdown

Purpose & Analysis:
Shows the step-by-step impact of different factors on profitability.

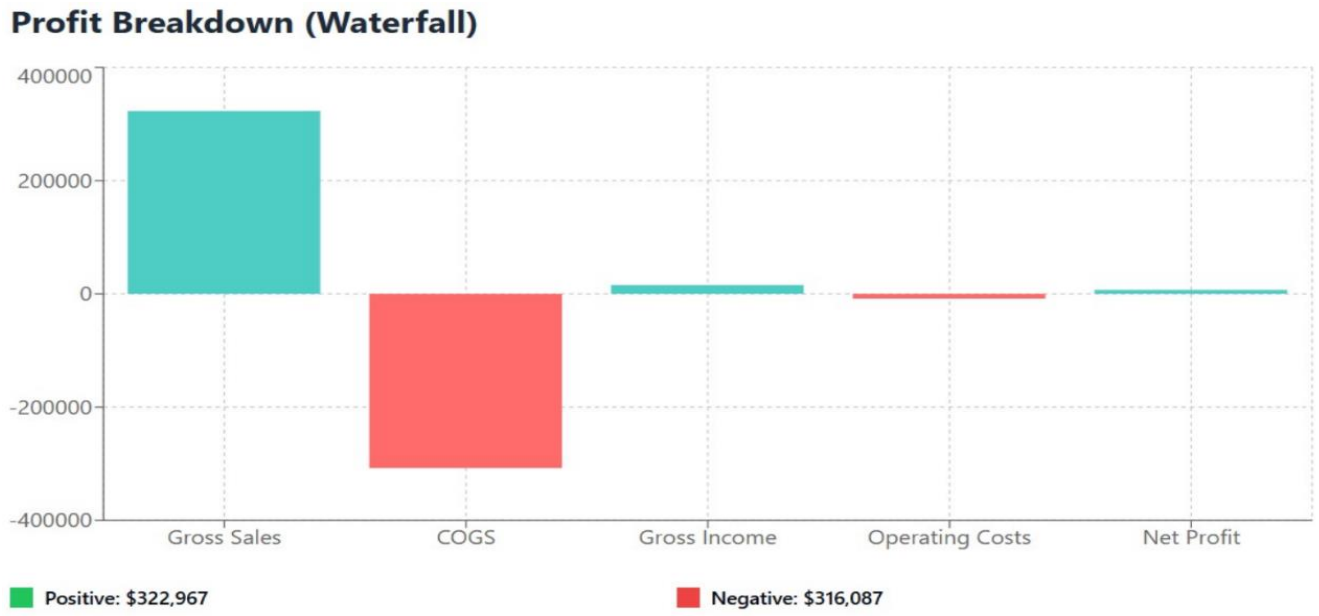
Flow:

Starting Point: Gross Sales Revenue

Deductions: COGS, operating expenses, taxes

Ending Point: Net Profit / Gross Income

Visual Flow:
Green bars represent gains, red bars represent deductions



Key Insights:

The **final profit margin** remains positive, indicating sustainable operations.

Cost of Goods Sold (COGS) has the largest negative impact on profit.

Despite deductions, **gross income remains a strong proportion of gross sales**, reflecting good cost control.

Overall Analysis & Recommendations

Summary of Key Findings:

Branch C (Naypyitaw) consistently performs best in terms of sales and profitability.

Food and Beverages is the top-selling product category across all branches.

Member customers contribute more revenue and show higher engagement.

Customer ratings are generally high, indicating good service quality and satisfaction.

Business Recommendations:

Strengthen membership programs with targeted offers to increase repeat purchases and customer loyalty.

Improve checkout and payment processes to reduce customer drop-offs during the purchase stage.

Focus inventory and promotions on high-performing product categories like Food & Beverages and Fashion Accessories.